PRANSHU KUMAR

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EDUCATION

Veermata Jijabai Technological Institute (VJTI), Mumbai **June 2027**
Bachelor of Technology in Electronics and Telecommunications **9.17 CGPA – Present**
Minor in Artificial Intelligence and Machine Learning

TECHNICAL SKILLS

- **Programming languages:** C, C++, Python, SQL
- **Embedded & Dev Tools:** nRF52840, ESP32, Zephyr RTOS, ESP-IDF, Arduino IDE, BLE, Git, pytest
- **ML, Vision & Data Analysis:** PyTorch, YOLOv8, OpenCV, EasyOCR, scikit-learn, NumPy, SciPy, Pandas, Excel
- **Simulation & CAD:** MATLAB/Simulink, ngspice, AutoCAD
- **Operating Systems:** Linux, Windows, macOS

PROFESSIONAL EXPERIENCE

IIT Bombay, Mumbai, India | *IoT Research Intern – Machine Intelligence Program* **June 2026 – July 2026**

- Independently conceived, pitched and built ParkinSense, a wearable continuous-monitoring platform for Parkinson's disease, from concept to a working end-to-end prototype (firmware, BLE protocol, real-time signal processing, dashboard) during the internship.
- Extended the lab's existing nRF52840/BLE IMU expertise into a three-pipeline architecture (motion, PPG/cardiac and activity) fusing inertial and optical sensing on one synchronised 104 Hz stream, moving beyond the single-modality tremor detection common in prior wearable PD research.
- Replaced offline, single-axis tremor analysis with real-time, multi-axis, motion-context-aware detection; replaced isolated activity tracking with a step counter cross-gated against tremor; and replaced single-channel monitoring with a motion-aware confidence model linking cardiac and activity readings back to tremor state.

Scientechnic, Dubai, UAE | *Project Engineering Intern – Building Technologies Department* **June 2025 – July 2025**

- Implemented, tested and commissioned a KNX-based lighting control system at Amazon DXB30 Innovation Hub, Dubai Internet City, including actuator configuration, addressing and line diagnostics on a central monitoring system.
- Designed initial I/O schedules (AI, DI, AO, DO) and conducted DDC selection for several BMS projects (e.g. Burj Binghatti Jacob & Co. Residences, GEMS Founders School), guided by equipment schedules, load demand, spare margins and peripheral device requirements.
- Assisted in professional AV system design tailored to client requirements (use case, venue acoustics, display sizing, lighting/BMS integration) across DSPs, matrix switchers, amplifiers, speakers and control processors.
- Evaluated AV-over-IP solutions for signal routing, network infrastructure and scalability in distributed AV environments.

IIT Bombay, Mumbai, India | *IoT Research Intern – Machine Intelligence Program* **Feb 2025 – May 2025**

- Conducted research on IoT sensor networks and real-time data transmission, acquiring high-speed IMU data on nRF52840 Sense and ESP32 and optimising BLE communication for efficient transfer.
- Collaborated on an industrial project with V-Guard collecting MPU data from underwater submerged pumps; investigated frequent EMI-related disruptions in the existing serial link over a waterproof cable.
- Contributed to transitioning the system from serial to UDP-based wireless transmission, improving data reliability and throughput under interference, and supported testing to ensure consistent real-time data retrieval in operation.

ACADEMIC PROJECTS

ParkinSense – Continuous Multi-Modal Wearable Platform for Parkinson's Monitoring (nRF52840, BLE, Signal Processing, Python)

- Architected a complete end-to-end wearable system (custom firmware, BLE protocol, three parallel real-time signal-processing pipelines, live dashboard) for continuous Parkinson's monitoring, addressing the gap left by single-modality tremor wearables that ignore cardiac and activity context.
- Built a 104 Hz sensor-fusion stream on a Seeed XIAO nRF52840 Sense with LSM6DS3 IMU and MAX30102 PPG, sending versioned 208-byte binary BLE packets (10 samples each) with sequence numbers, gap detection and measured device rate; designed BLE protocol v2 for $\sim$100 Hz PPG.
- Designed the tremor pipeline: Welch PSD on 4 s windows every 0.5 s, 3.5–7.5 Hz peak search with interpolation, SNR and harmonic rejection, REST/LOW MOTION/ACTIVE motion context and a 2-state HMM; raised sensitivity from 0.69 to 0.99 at 1.00 specificity with 0.024 Hz median frequency error (synthetic ground truth).

- Developed a PPG pipeline from scratch (finger detection, signal-quality index, adaptive peak detection, RR intervals) producing heart rate, HRV (RMSSD, SDNN, pNN50) and SpO_2 calibrated against an Amazfit Helio Strap, with SEARCHING, LOCKED, TRACKING and LOST lock states; HR MAE at rest 0.37 BPM (synthetic).
- Engineered an accelerometer step counter gated by motion context and a tremor guard, cutting false steps during rest tremor from 23.3 to 0.3/min (synthetic); counted 69 steps against ≈86 expected on a real walking recording.
- Built a FastAPI + Plotly Dash runtime with bit-exact replay, per-recording provenance (79-column capture) and an 84-test CI suite with golden regression on real recordings, enabling offline analysis and future ML dataset generation.

TremoSense – Anti-Tremor Assistive Spoon for Parkinson's Patients (Zephyr RTOS, C, EKF, LQR, Simulink)

- Engineered a 2-axis active stabilisation system that suppresses pathological hand tremor while preserving intentional motion, improving usability for Parkinson's patients during eating.
- Integrated the LSM6DS3 6-DoF IMU on a XIAO nRF52840 Sense (ARM Cortex-M4F) sampling at 208 Hz, driving an MG995 roll and FS90MG pitch servo with commands every 10 ms through Zephyr threads, a message queue and a semaphore.
- Designed a 2-state Extended Kalman Filter for roll/pitch estimation, removing gyroscope drift and high-frequency noise, and a 2×4 LQR gain for coordinated two-axis stabilisation with lower control effort than PID.
- Validated the EKF–LQR design in MATLAB/Simulink at 1000 Hz: over 90% roll and 80% pitch suppression across most of the 3–7 Hz tremor band; residual RMS 0.346°/0.573° under 8°, 8 Hz tremor.

NaviSight – Assistive Navigation for the Visually Impaired (Arduino, ESP32-CAM, TensorFlow Lite)

- Developed a dual-component assistive system: a smart cane with obstacle detection and an ESP32-CAM vision module with real-time object recognition, for independent navigation, object awareness and personal safety.

1. *Echo-Stride – Smart Cane*

- Engineered a smart cane with ultrasonic sensors on an Arduino Uno for obstacle detection, with haptic (vibration motor) and buzzer alerts and an ergonomic, durable casing for daily use.
- Added GPS (L80-M39) and Bluetooth (HC-05) to send the user's location to their phone, and configured Tasker automation to share the live Google Maps location with trusted contacts by SMS at set intervals.

2. *EyeCue – ESP32-CAM Smart Vision Module*

- Trained and deployed lightweight TensorFlow Lite models on the ESP32-CAM for offline, low-power object/person recognition, with on-device preprocessing and inference.
- Integrating text-to-speech audio feedback and optimising form factor and power for daily outdoor use.

CircuitVision – Hand-Drawn Circuit to SPICE Netlist (AI/ML minor, team of 2; YOLOv8, OpenCV, EasyOCR, ngspice)

- Built an end-to-end pipeline that turns a photo of a hand-drawn circuit into a simulated SPICE netlist: binarisation, symbol detection, orientation, wire tracing, value reading, netlist generation, grammar validation and ngspice simulation.
- Trained YOLOv8s on the CGHD handwritten-circuit dataset (2,981 training photos, 9 classes), reaching 0.904 mAP50 and 0.611 mAP50-95 on a held-out drafter.
- Implemented paper-relative binarisation, HOG + SVM orientation and polarity classification, and wire tracing with symbol masking and crossover handling (0.994 exact topology across 11 synthetic photo conditions).
- Generated netlists from EasyOCR-read values, validated by a CFG/Earley parser before simulation; added an HMM (Viterbi) text-to-netlist path scoring 0.90 exact on held-out sentences; delivered a Gradio app and 136 automated tests.
- Measured honestly on real photos: 37.5% of 48 analog circuits fully correct with ground-truth boxes, 10.4% with detector boxes; identified handwritten-value OCR as the main bottleneck.

Wall-E – Self-Balancing, Line-Following Robot (ESP32, ESP-IDF, Git; SRA)

- Co-designed a two-wheeled self-balancing, line-following robot on an ESP32 with the SRA's in-house SRA Board.
- Integrated a Light Sensor Array to detect white or black lines from reflectivity differences, and implemented PID control for line tracking and for balancing from positional feedback.
- Tuned PID gains through iterative testing to achieve reliable line-following and self-balancing across test scenarios.

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Academic Branch Topper (First Year).
- Organising team, Technovanza 2024, Asia's second-largest tech fest (4,000–5,000 participants).
- Member, Society of Robotics and Automation (SRA), VJTI; active in workshops.
- Guitarist at college events and cultural programmes.